AIML PYTHON MINI PROJECT SUBMISSION

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Domain/subject name: AIML (Artificial Intelligence & Machine Learning)

College and Branch name: Government Polytechnic College Washim (IF)

Project 1: Student Management System

Code :-

```
students = {}




# Add Student

def add_student():

    try:

        roll = int(input("Enter Roll Number: "))


        if roll in students:

            print(" Roll Number Already Exists!")

            return


        name = input("Enter Name: ")


        marks = []


        for i in range(1, 6):

            mark = float(input(f"Enter Subject {i} Marks: "))
```

```
            marks.append(mark)

        percentage = sum(marks) / 5
        grade = calculate_grade(percentage)

        students[roll] = {
            "name": name,
            "marks": marks,
            "percentage": percentage,
            "grade": grade
        }

        print(" Student Added Successfully!")

    except ValueError:
        print("Invalid Input!")

# Grade Calculation
def calculate_grade(percentage):
    if percentage >= 90:
        return "A+"
    elif percentage >= 80:
```

```
        return "A"
    elif percentage >= 70:
        return "B+"
    elif percentage >= 60:
        return "B"
    elif percentage >= 50:
        return "C"
    else:
        return "F"


# View All Students
def view_all():
    if not students:
        print("No Records Found!")
        return

    sorted_students = sorted(
        students.items(),
        key=lambda x: x[1]["percentage"],
        reverse=True
    )
```

```
    print("\n" + "=" * 75)
    print(f"{'Rank':<6}{'Roll':<10}{'Name':<20}{'%':<15}{'Grade'}")
    print("=" * 75)

    rank = 1

    for roll, data in sorted_students:
        print(
            f"{rank:<6}{roll:<10}{data['name']:<20}"
            f"{data['percentage']:<15.2f}{data['grade']}"
        )
        rank += 1

    print("=" * 75)

# Search Student
def search_student():
    try:
        roll = int(input("Enter Roll Number: "))

        if roll in students:
```

```
            data = students[roll]


            print("\nStudent Details")

            print("-" * 30)

            print("Roll Number :", roll)

            print("Name         :", data["name"])

            print("Marks        :", data["marks"])

            print("Percentage  :", round(data["percentage"], 2))

            print("Grade        :", data["grade"])


        else:

            print(" Student Not Found!")


    except ValueError:

        print(" Invalid Roll Number!")


# Update Student

def update_student():

    try:

        roll = int(input("Enter Roll Number: "))


        if roll in students:
```

```
        marks = []

        for i in range(1, 6):
            mark = float(input(f"Enter New Subject {i} Marks: "))
            marks.append(mark)

        percentage = sum(marks) / 5
        grade = calculate_grade(percentage)

        students[roll]["marks"] = marks
        students[roll]["percentage"] = percentage
        students[roll]["grade"] = grade

        print(" Record Updated Successfully!")

    else:
        print(" Student Not Found!")

except ValueError:
    print(" Invalid Input!")
```

```python
# Delete Student
def delete_student():
    try:
        roll = int(input("Enter Roll Number: "))

        if roll in students:

            confirm = input(
                "Are you sure? (yes/no): "
            ).lower()

            if confirm == "yes":
                del students[roll]
                print("Record Deleted Successfully!")

        else:
            print(" Student Not Found!")

    except ValueError:
        print(" Invalid Roll Number!")


# Class Report
```

```
def class_report():


    if not students:
        print("No Records Found!")
        return


    topper = max(
        students.items(),
        key=lambda x: x[1]["percentage"]
    )


    average = (
        sum(student["percentage"]
            for student in students.values())
        / len(students)
    )


    pass_count  =0
    fail_count = 0


    for student in students.values():
        if student["percentage"] >= 50:
```

```
            pass_count += 1
        else:
            fail_count += 1


    print("\n===== CLASS REPORT =====")
    print("Topper :", topper[1]["name"])
    print("Top Percentage :", round(topper[1]["percentage"], 2))
    print("Average Percentage :", round(average, 2))
    print("Pass Students :", pass_count)
    print("Fail Students :", fail_count)


# Rank List
def show_rank_list():


    if not students:
        print("No Records Found!")
        return


    sorted_students = sorted(
        students.items(),
        key=lambda x: x[1]["percentage"],
        reverse=True
```

```
    )


    print("\n===== RANK LIST =====")


    rank = 1


    for roll, data in sorted_students:


        print(
            f"Rank {rank} | "
            f"Roll: {roll} | "
            f"Name: {data['name']} | "
            f"Percentage: {data['percentage']:.2f}"
        )


        rank += 1



# Main Menu
while True:


    print("\n===== STUDENT MANAGEMENT SYSTEM =====")
```

```
print("1. Add Student")

print("2. View All Students")

print("3. Search Student")

print("4. Update Student")

print("5. Delete Student")

print("6. Class Report")

print("7. Rank List")

print("8. Exit")


choice = input("Enter Choice: ")


if choice == "1":

    add_student()


elif choice == "2":

    view_all()


elif choice == "3":

    search_student()


elif choice == "4":

    update_student()
```

```
elif choice == "5":
    delete_student()

elif choice == "6":
    class_report()

elif choice == "7":
    show_rank_list()

else:
    print(" Invalid Choice!")
```

Output Screenshort :-

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   TEST RESULTS   PORTS                Python: student_management

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe c:/Users/sdf/Desktop/ITR2026/Projects/student_management.py
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 2

===========================================================================
Rank  Roll      Name                %              Grade
===========================================================================
1     549       Abhishekh Jadhav    89.60          A
===========================================================================
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   TEST RESULTS   PORTS                Python: student_management

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe c:/Users/sdf/Desktop/ITR2026/Projects/student_management.py
===========================================================================

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 3
Enter Roll Number: 549

Student Details
------------------------------
Roll Number : 549
Name        : Abhishekh Jadhav
Marks       : [98.0, 89.0, 96.0, 76.0, 89.0]
Percentage  : 89.6
Grade       : A

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 4
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   TEST RESULTS   PORTS                Python: student_management

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe c:/Users/sdf/Desktop/ITR2026/Projects/student_management.py
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 4
Enter Roll Number: 549
Enter New Subject 1 Marks: 98
Enter New Subject 2 Marks: 97
Enter New Subject 3 Marks: 91
Enter New Subject 4 Marks: 89
Enter New Subject 5 Marks: 100
 Record Updated Successfully!
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    TEST RESULTS    PORTS          Python: student_management

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe c:/Users/sdf/Desktop/ITR2026/Projects/student_management.py

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 6

===== CLASS REPORT =====
Topper : Abhishekh Jadhav
Top Percentage : 95.0
Average Percentage : 95.0
Pass Students : 1
Fail Students : 0
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    TEST RESULTS    PORTS          Python: student_management

> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe c:/Users/sdf/Desktop/ITR2026/Projects/student_management.py

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 7

===== RANK LIST =====
Rank 1 | Roll: 549 | Name: Abhishekh Jadhav | Percentage: 95.00

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 5
Enter Roll Number: 549
Are you sure? (yes/no): yes
Record Deleted Successfully!
```

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 6

===== CLASS REPORT =====
Topper : Abhishekh Jadhav
Top Percentage : 95.0
Average Percentage : 95.0
Pass Students : 1
Fail Students : 0
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL TEST RESULTS PORTS Python: student_management

```
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe c:/Users/sdf/Desktop/ITR2026/Projects/student_management.py
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 7

===== RANK LIST =====
Rank 1 | Roll: 549 | Name: Abhishekh Jadhav | Percentage: 95.00

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 5
Enter Roll Number: 549
Are you sure? (yes/no): yes
Record Deleted Successfully!
```

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice: 5
Enter Roll Number: 549
Are you sure? (yes/no): yes
Record Deleted Successfully!

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Report
7. Rank List
8. Exit
Enter Choice:
```